वाणी सेतु · a bridge of voice

VaaniSetu

A multilingual AI document reader on Google Cloud

Vertex AI Gemini · Cloud Vision · Text-to-Speech · Cloud Run functions · API Gateway · Cloud Storage · Firestore



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Cloud Application Development (CAP 303) · TY B.Tech Sem V · uGDX School of Technology · AY 2026-27

The problem

Everyday life runs on paper, but not everyone can read it comfortably



Low vision & elderly

Small print on prescriptions and bills is hard to read, and mistakes with doses or due dates have real consequences.



Limited literacy

Official notices and forms use formal, technical language that is hard to understand even when it can be read.



Language barrier

Migrants and students often receive documents in a language they do not read, such as a Hindi bill for a non-Hindi reader.

Our solution: VaaniSetu

Photograph a document and hear it explained in your language in about 10 seconds

Snap

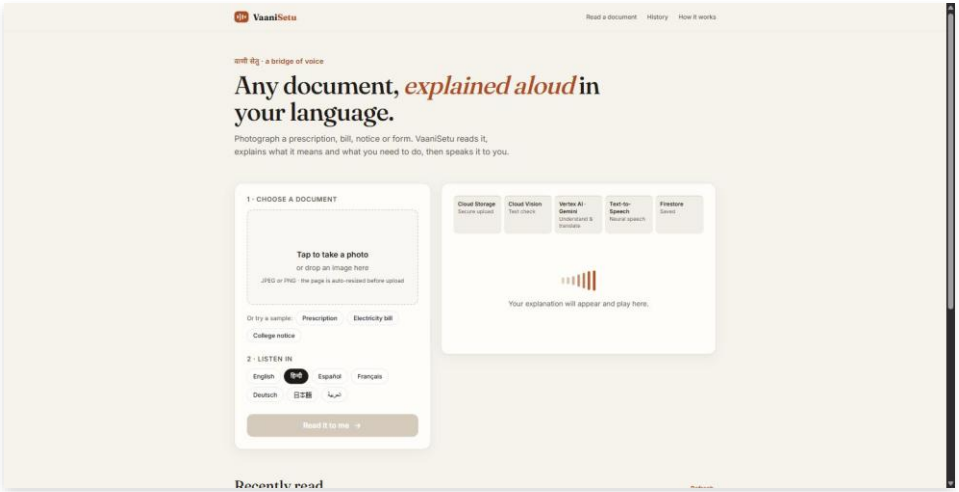
Take a photo of a prescription, bill, notice or form

Understand

AI reads it and explains what it means and what to do

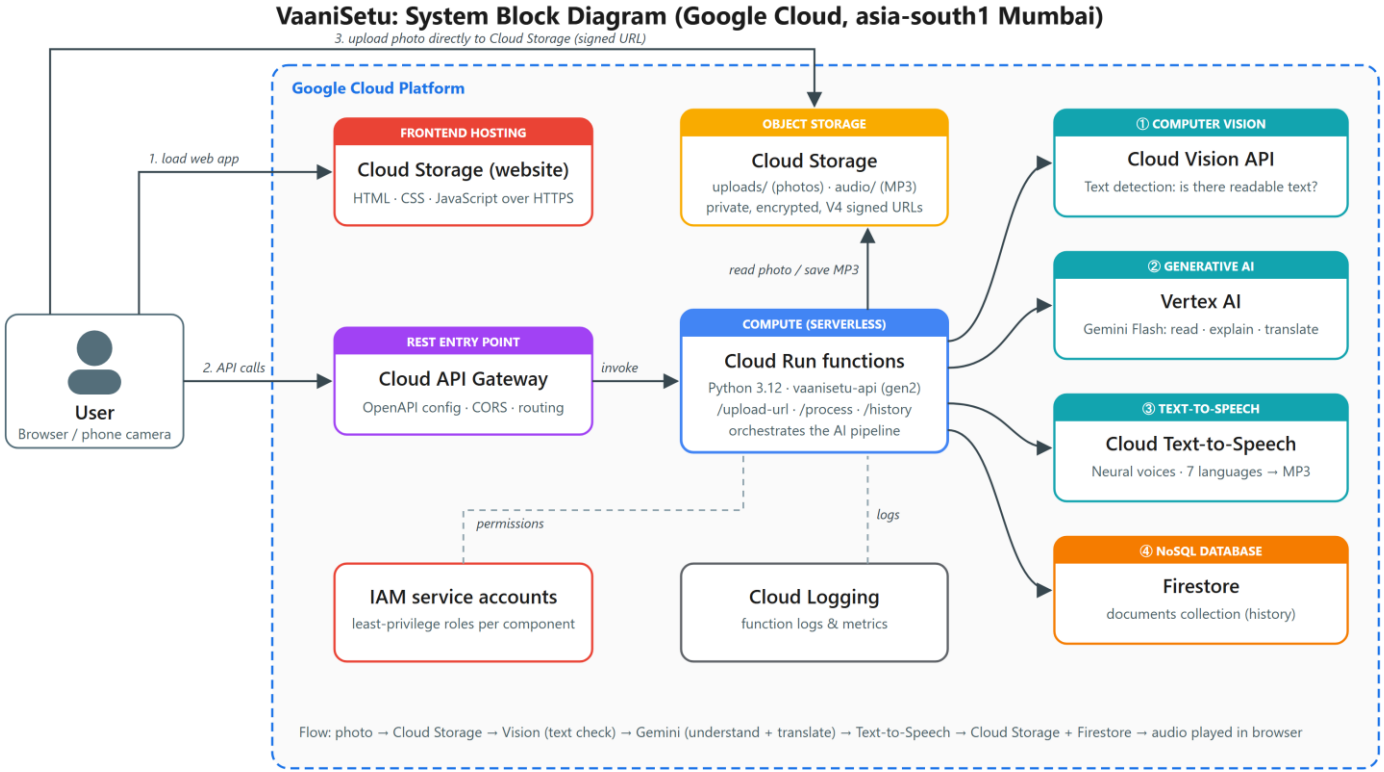
Listen

A natural neural voice reads the explanation aloud



7 output languages · any input language · key points & action items · history

Block diagram



How a request flows

One photo, five Google Cloud services, about 11 seconds

1

Cloud Storage

Signed upload

Browser uploads the photo directly to a private bucket

2

Cloud Vision

Text check

Text detection rejects photos with no readable text

3

Vertex AI

Understand

Gemini transcribes, explains and translates

4

Text-to-Speech

Speak

Neural voice produces an MP3, saved to Cloud Storage

5

Firestore

Remember

Result stored and shown in history

API Gateway → Cloud Run function orchestrates every step · least-privilege service accounts · Cloud Logging

Cloud services used

8 core Google Cloud services plus IAM and Cloud Logging, all serverless or fully managed



Cloud Storage

Photos & MP3 audio, private + encrypted



Cloud Run

Python 3.12 function orchestrating the AI pipeline



API Gateway

OpenAPI routes: /upload-url, /process, /history



Vertex AI

Gemini 3.5 Flash-Lite reads, explains, translates



Cloud Vision

Text detection gate before Gemini runs



Text-to-Speech

Neural voices in 7 languages



Firestore

Serverless NoSQL history (free tier)



Storage hosting

HTTPS static website for the web app



IAM

Least-privilege service accounts



Cloud Logging

Logs & metrics for every request

Development technologies

Python 3.12 + Google Cloud SDKs

Cloud Run function calling Storage, Vision, Vertex AI (google-genai), Text-to-Speech and Firestore

HTML5

Semantic, accessible page structure with a skip link and ARIA roles

CSS3

Responsive grid, design tokens, animated pipeline tracker, reduced-motion support

JavaScript (ES2020)

Camera / drag-and-drop upload, Canvas resizing, Fetch API, audio playback

gcloud CLI + Bash

One repeatable script, run in Cloud Shell, provisions every resource

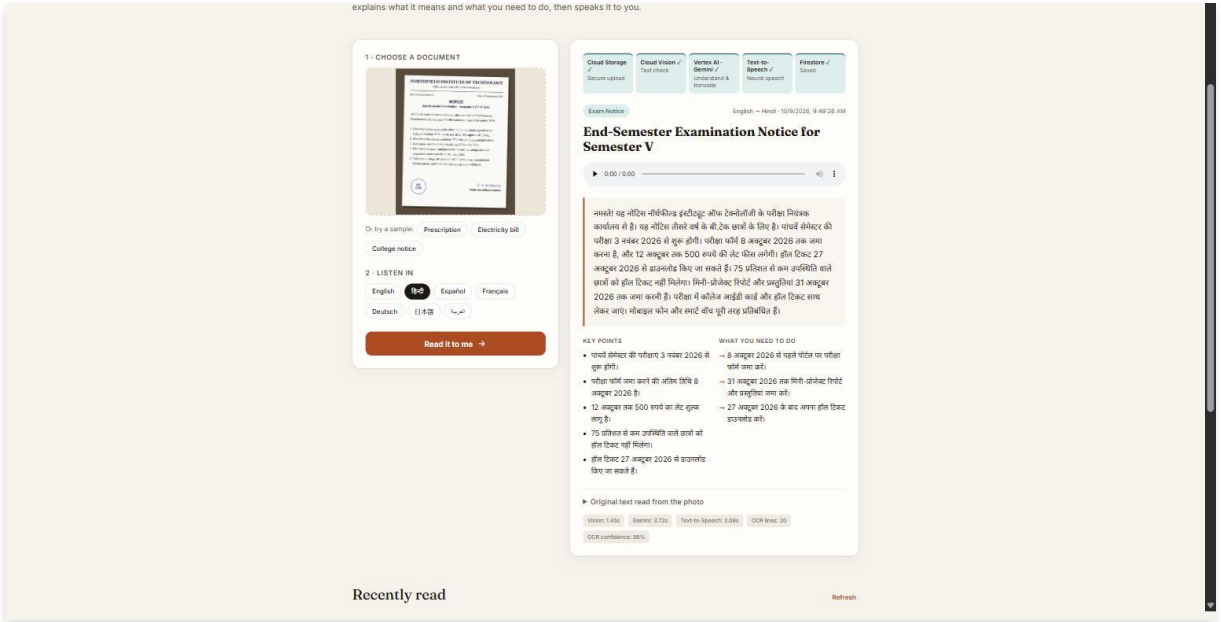
Git + GitHub

Version control with small, atomic commits

JSON

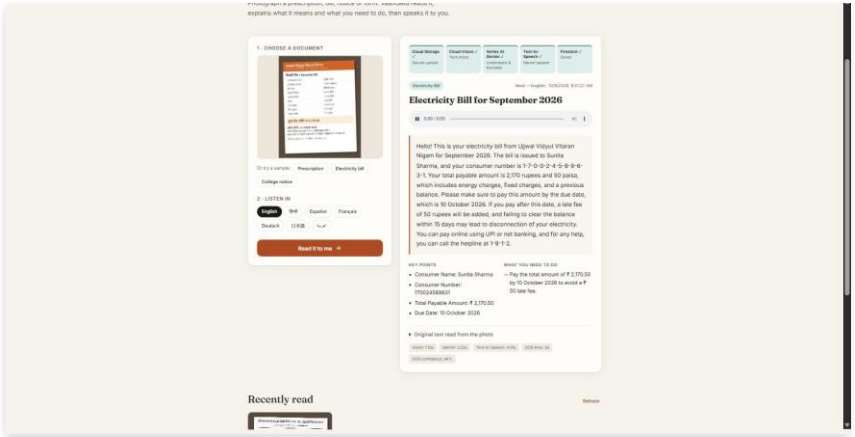
API payloads, structured AI output and the OpenAPI spec

Demo: English notice → Hindi speech

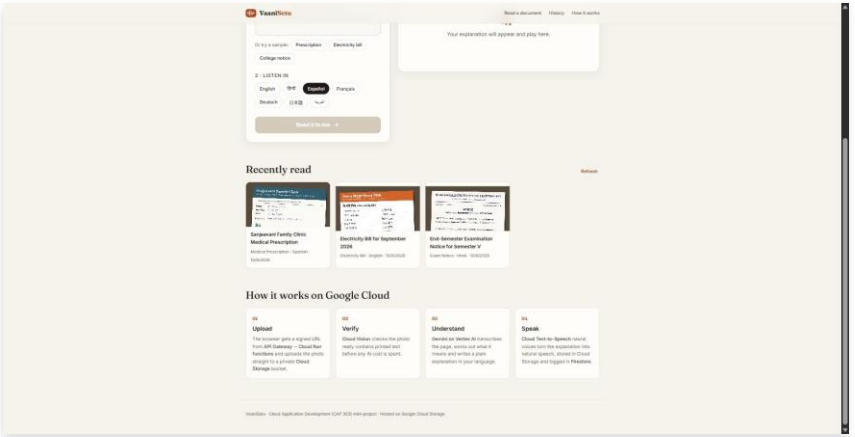


- Document type detected: exam notice
- Summary written in Hindi and spoken by a neural Text-to-Speech voice
- Key points and action items with deadlines
- All five pipeline stages complete
- Cloud Vision OCR confidence 98%

Demo: Hindi bill → English, and history

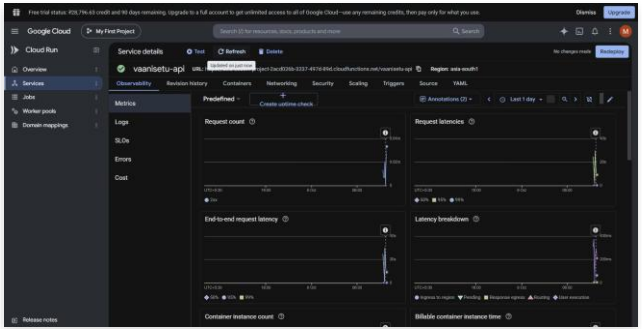


A bill printed in Hindi is explained in English: amount due ₹2,170.50, due date 10 October 2026, late fee ₹50.

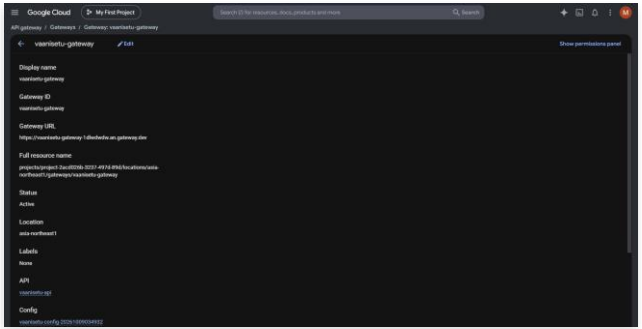


Every result is saved in Firestore. Tapping a card replays the explanation and audio.

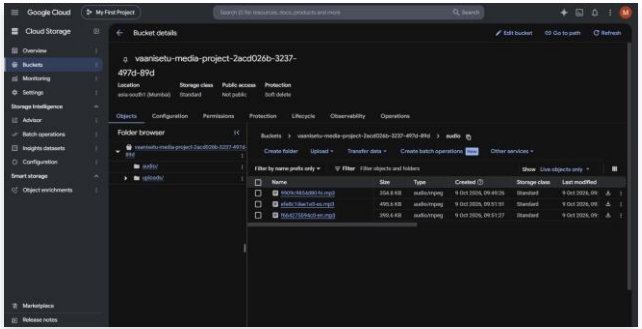
Deployed on Google Cloud (asia-south1)



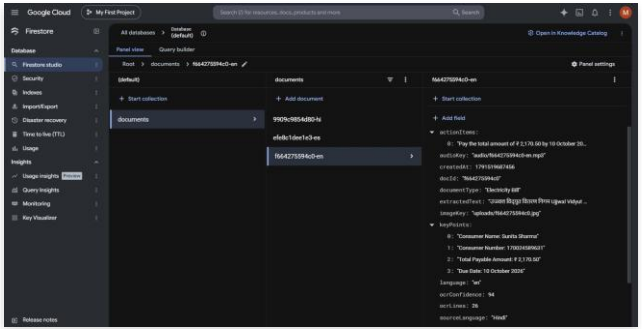
Cloud Run function metrics



API Gateway: active



Cloud Storage: MP3 files



Firestore record

Performance at a glance

Measured per document during testing

~1.5 s
Cloud Vision text check

~3.5 s
Gemini understanding

~2 s
Text-to-Speech

~11 s
Photo to speech, end to end

7 output languages

10 Google Cloud services

₹0 cost while idle

Security & cost by design

Private storage

Public access prevention enforced, encrypted at rest, uploads deleted after 30 days

Signed URLs

Browser gets 5-minute upload and 1-hour download links, never Google credentials

Least privilege

Separate service accounts for function, gateway and builds, each with only the roles it needs

Spend guard

At most 3 function instances; ₹400 budget alert at 50/90/100%

Cost gate

Cloud Vision blocks non-document photos before any Gemini charge

Scales to zero

No servers to patch; nothing runs and nothing bills while idle

Challenges solved

Challenge	How it was solved
The AWS account was suspended mid-project	Rebuilt on Google Cloud with one re-runnable deploy script; AWS version kept in Git history
Gemini 3.8 Flash took 16-24 s per document	Benchmarked 6 models: 3.5 Flash-Lite with thinking off takes ~3.5 s and costs less
API Gateway is not offered in Mumbai	Gateway runs in Tokyo (asia-northeast1) and routes to the function in Mumbai
Cloud Run has no private key to sign URLs	The function signs Storage URLs via the IAM signBlob API with its own service account

Conclusion & future scope

What was achieved

- A working, deployed serverless app that turns any document photo into spoken guidance
- 8 core Google Cloud services + IAM and Cloud Logging, integrated end to end
- Secure, cheap to run, and deployed with one script

What comes next

- Multi-page PDFs with Pub/Sub and Workflows
- More Indian-language voices and a spoken Q&A mode
- Private per-user history with Firebase Authentication
- WhatsApp / IVR channel for feature phones

Live: storage.googleapis.com/vaanisetu-web-project-2acd026b-3237-497d-89d/index.html · Code: github.com/17mohak/vaanisetu

Thank you

Questions? · प्रश्न?